Stanford

Weekly update 3 of 5 · week of Nov 23

Prompt v2: adding qualitative knowledge

THIS UPDATE IN THREE NUMBERS

13.8%

run-to-run CV (v1 19.0%): more consistent

15.1%

coverage (v1 19.5%): less accurate

89%

of v2 answers still assume polyiso roof insulation

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What prompt v2 changed

1 Scope

Written to match the dataset's enclosure definition.

2 Component take-off

Areas → assemblies → intensities → total.

3 High-carbon materials

HFC-blown XPS "ten times or more" the GWP of other insulation per unit R; aluminum curtain wall 100–200 $kgCO_2e/m^2$.

4 No anchoring

Estimating from a "typical building" is ruled out.

5 Renovation rule

Count only the materials of the envelope work.

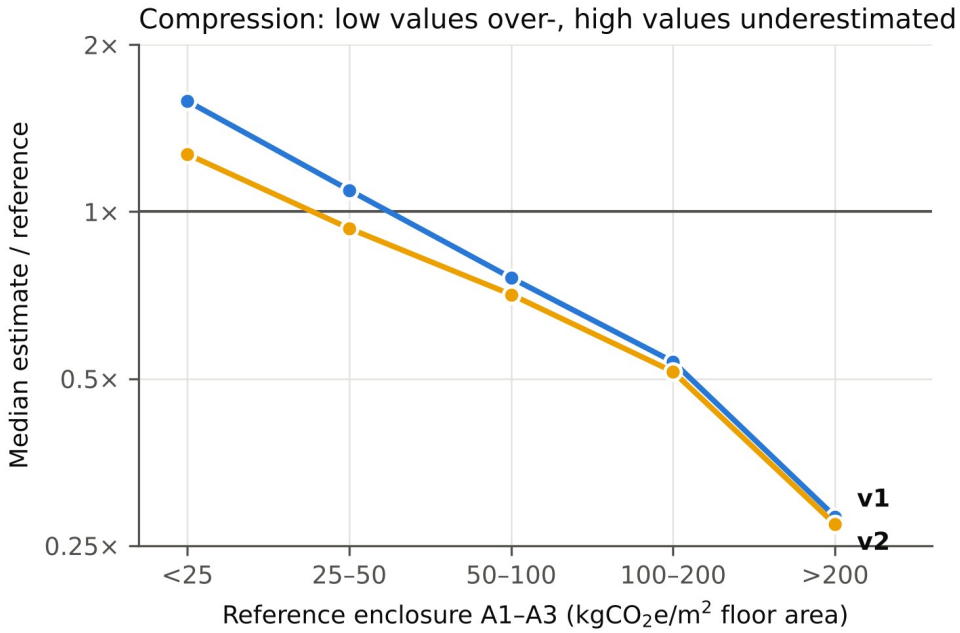
6 BEST value

A best estimate is requested alongside the range.

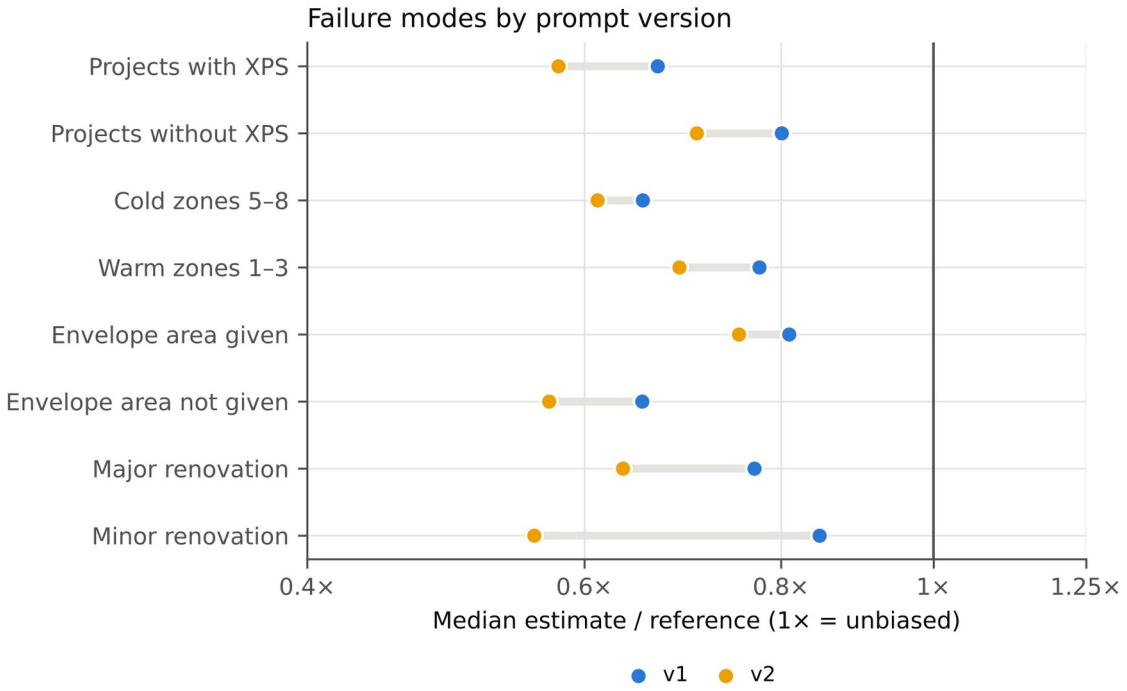
Results: v2 vs v1 (292 buildings, 3 runs each)

Prompt	Coverage	RMSE	Med. APE	Med. |log|	Bias	CV
v1	19.5%	79.7	37.9%	0.456	0.75	19.0%
v2	15.1%	81.0	44.3%	0.563	0.67	13.8%

- v2 errors are **significantly larger** (Wilcoxon p < 0.0001).
- Compression eased (slope 0.46 → 0.55), but the **whole curve shifted down**.
- More consistent across runs: CV 19.0% → 13.8%.



Why v2 got worse



1. **The facts were repeated, not applied.** XPS was named 50× more often, but 89% of answers still assumed polyiso roofs; the median roof intensity stayed at 35 $kgCO_2e/m^2$.
2. **The take-offs were too light.** Bottom-up intensities came out at 66.8 per m^2 of envelope, against 86.7 in the reference.
3. **A subtracting rule hurt.** "Count only new materials" dropped minor renovations from 0.85 to 0.56.

Lesson for prompt design

Qualitative knowledge gets repeated, not applied. To move the numbers, give numbers, at the step where the model chooses them.

This is the design principle behind **prompt v3**.

Next update

Coming next

- **Prompt v3:** sourced quantitative anchors. A table of external A1–A3 intensities per m² of component, placed at the decision step.
- **Held-out check** on buildings not used for the prompt diagnostics.

Input we'd like

- Sources for intensities: CLF baselines, PCI and Kawneer EPDs, RDH studies. **None from the ground-truth dataset or the CLF WBLCA benchmark** (leakage). Any to add or avoid?